STAT 6611
Fall 2026 - Nordman
Midterm Exam
10/15/26

STUDENT NAME (PRINTED):_____

I have neither given nor received any unauthorized aid on this exam.

Student Signature:_____ Date:_____

- The exam is due **Friday, October 23 by midnight in gradescope.**
- Students may use notes but must work independently and alone.
- Students may not use AI.
- Students are not to discuss the exam with one another during the exam process.
- Be sure to show work to receive credit.
- The instructor may choose a subset of this exam for a 10 minute closed-book follow-up quiz, administered on Tuesday, Oct 27 in class.
- Exams are always stressful, but you can do it. GOOD LUCK!

1. Let A be an $n \times m$ matrix of rank r.

   (a) Show that $\text{rank}(I - AA^-) = n - r$.

   (b) Find the rank of $I - A^-A$.

2. Suppose that $\underline{y}$ is an $n \times 1$ observable random vector that follows the model

$$\underline{y} = X(\underline{\beta} + \underline{b}) + \underline{e},$$

   where X is an $n \times p$ known matrix, and $\underline{\beta}$ is a $p \times 1$ vector of unknown parameters. Furthermore, suppose that $\underline{b}$ and $\underline{e}$ are independent random vectors with $\underline{b} \sim N(\underline{0}, D)$ and $\underline{e} \sim N(\underline{0}, \sigma^2 I)$, where D is an unknown non-negative definite matrix, and σ^2 is an unknown positive parameter.

   (a) Show that the simple least squares estimator (SLSE) of any estimable function $\underline{\lambda}'\underline{\beta}$ is the best linear unbiased estimator (BLUE) of $\underline{\lambda}'\underline{\beta}$.

   (b) Show that the variance of the BLUE of any estimable function $\underline{\lambda}'\underline{\beta}$ is $\sigma^2\underline{\lambda}'(X'X)^-\underline{\lambda} + \underline{\lambda}'D\underline{\lambda}$.

3. Suppose that $\underline{y}$ is an $n \times 1$ observable random vector that follows the linear model

$$\underline{y} = X\underline{\beta} + \underline{\epsilon},$$

   where $\underline{\beta}$ is a $p \times 1$ vector of unknown parameters, $\underline{\epsilon}$ is an unobservable random vector whose distribution is $N(\underline{0}, \sigma^2 I)$, and σ^2 is an unknown positive parameter. Further, let $X\underline{\beta} = X_1\underline{\beta}_1 + X_2\underline{\beta}_2$, where $X = (X_1, X_2)$, $\underline{\beta} = (\underline{\beta}_1', \underline{\beta}_2')'$, and $\underline{\beta}_1$ is a $q \times 1$ $(1 \leq q < p)$ vector. Suppose that the parameters $\underline{\beta}$ satisfy the constraints $\underline{\beta}_2 = \underline{b}$. That is, we now have Model IIa.

   (a) Let $\underline{\lambda}'\underline{\beta} = \underline{\lambda}_1'\underline{\beta}_1 + \underline{\lambda}_2'\underline{\beta}_2$ be a linear function. Show that $\underline{\lambda}'\underline{\beta}$ is estimable if and only if $\underline{\lambda}_1' \in \mathcal{R}(X_1)$, where $\mathcal{R}(X_1)$ denotes the row space of X_1.

   (b) Let $\underline{\lambda}'\underline{\beta} = \underline{\lambda}_1'\underline{\beta}_1 + \underline{\lambda}_2'\underline{\beta}_2$ be an estimable function. Show that the BLUE of $\underline{\lambda}'\underline{\beta}$ is given by

$$\underline{\lambda}_1'(X_1'X_1)^-X_1'(\underline{y} - X_2\underline{b}) + \underline{\lambda}_2'\underline{b}.$$

4. Consider the following linear model

$$y_i = \mu + \alpha x_i + \gamma z_i + \epsilon_i, \quad i = 1, \ldots, n, \tag{1}$$

   where μ, α, and γ are unknown fixed parameters; the y_i's are observable random variables; the x_i's and z_i's are known constants; the ϵ_i's are unobservable and uncorrelated random variables with mean 0 and variance σ^2, where σ^2 is an unknown positive parameter. Further,
   let $\underline{1} = (1, \ldots, 1)'$ be an $n \times 1$ vector of 1's, $\underline{y} = (y_1, \ldots, y_n)'$, $\underline{x} = (x_1, \ldots, x_n)'$, and $\underline{z} = (z_1, \ldots, z_n)'$. Suppose also that $\underline{1}'\underline{x} = \underline{1}'\underline{z} = \underline{x}'\underline{z} = 0$ and $\underline{x}'\underline{x} = \underline{z}'\underline{z} = n$.

   (a) Formulate model (1) as a Model II: $\underline{y} = X\underline{\beta} + \underline{\epsilon}$. Carefully write out $X, \underline{\beta}$, and $\underline{\epsilon}$.

   (b) Calculate $X'X$ and use this to argue that $n \geq 3$.

   (c) Show that μ, α, and γ are estimable functions.

   (d) Find the best linear unbiased estimators (BLUE's) of μ, α, and γ in non-matrix, explicit expressions.

5. Consider model (1) with the constraint $\gamma = \mu + \alpha$. That is, we now have a constrained Model IIa. Let $\underline{\beta} = \begin{pmatrix} \mu \\ \alpha \\ \gamma \end{pmatrix}$ and $\underline{\lambda}' = (\lambda_1, \lambda_2, \lambda_3)$.

   (a) Find all estimable functions $\underline{\lambda}'\underline{\beta}$.

   (b) Give the constrained conjugate normal equations, in an explicit matrix form, for an estimable $\underline{\lambda}'\underline{\beta}$.

   (c) Give the BLUE of any an estimable function $\underline{\lambda}'\underline{\beta}$ in a non-matrix, explicit expression.